In addition to its theoretical significance, the study of organizations has much practical significance today, particularly in a democratic society. While large bureaucracies have existed for thousands of years, it has been primarily in this century that they have expanded greatly and come to pervade all of social life. Since formalized organizations have become the dominant form of institution in modern societies, a thorough knowledge of them is essential for an understanding of contemporary social life. The centralization of power in the hands of management that organizational giants make possible, moreover, poses a challenge to democracies. An efficient administrative machinery vests tremendous power in the hands of the few—be they corporation managers, government officials, military officers, party bosses, or union leaders—and thereby undermines the sovereignty of the many to whom the few in a democracy are expected to be responsible. Acquiring knowledge about bureaucratic organizations is an important first step in meeting the threat they pose for democratic institutions.[12]

Methods in the Study of Organizations

Methods of Data Collection. The methods of social research can be classified in various ways. One classificatory scheme emphasizes the purpose for which the data are collected.[13] The types arrived at include (1) exploratory studies, in which familiarity is gained with some problem or new insights are achieved that can guide further research; (2) descriptive studies, which define and portray the characteristics of the object of research or determine the frequency of various occurrences and examine their associations with one another; and (3) hypothesis-testing studies, which focus on the collection of data that permit the confirmation of a given hypothesis, or set of hypotheses, and thereby help to determine the probable validity of the theory from which it is derived. Needless to say, many studies have a mixed purpose, as exemplified by a recent work in the area of formal organization—a case study of the International Typographical Union.[14] This

[12] Lipset suggests that while the great social theorists of the nineteenth century directed their attention to the study of the sources of conflict and consensus—these themes dominating the work of both Marx and Tocqueville—the major social scientists of the early twentieth century have been much more concerned with the problems posed for democracy by the growth of bureaucratic institutional arrangements, as evidenced particularly in the writings of Michels and Weber. See Seymour M. Lipset, *Political Man,* Garden City, N.Y.: Doubleday, 1960, pp. 28–30.

[13] Claire Selltiz *et al., Research Methods in Social Relations* (2d ed.), New York: Holt, 1959, pp. 50–51.

[14] Seymour Lipset *et al., Union Democracy,* Glencoe, Ill.: Free Press, 1956.

study is descriptive when the characteristics and history of the ITU are recounted; it is hypothesis-testing when specific propositions about the occupational community of printers and its relation to union politics are tested; and it becomes exploratory when an attempt is made to formulate propositions "aimed at identifying the factors which make for and sustain democracy in private organizations."[15]

Research methods may also be classified according to the techniques employed in the collection of data. In principle, there are three ways of obtaining information about people: by watching them, by asking them questions, and by examining materials that have been written by them or about them. The three categories of research techniques which correspond to these operations are: observation, interviewing, and the analysis of documents. One group of these techniques may be used to the exclusion of the others, or a study may combine all three approaches. Besides, each of these techniques may be employed more or less rigorously. The use of these three methods of data collection in organizational research will be briefly discussed.

Considering first observation, the operation of watching people may be unstructured in the sense that the observer attempts to keep as complete a record as possible of all the activities of the members of the group being studied. Such an unstructured technique was utilized by the observer placed in the now famous Hawthorne Bank Wiring Observation Room.[16] While unstructured observation is still often used, particularly in exploratory research, "the movement has been away from exhaustive recording of all the behavior which occurs toward the selection of particular aspects of behavior in order to answer specific questions."[17] The observer who utilizes systematic observation methods either classifies the behavior observed into a set of predetermined categories or assigns to it a numerical index. Two widely used techniques of systematic observation, both concerned with recording patterns of social interaction, are those developed by Bales[18] and by Chapple.[19] In addition to these direct methods of observation, there is the use of

[15] *Ibid.*, p. 413.

[16] Roethlisberger and Dickson, *op. cit.*, pp. 387–391.

[17] Roger W. Heyns and Ronald Lippitt, "Systematic Observational Techniques," Gardner Lindzey (ed.), *Handbook of Social Psychology*, Cambridge, Mass.: Addison-Wesley, 1954, vol. I, p. 371. This essay contains probably the best survey of systematic observational techniques to be found in the literature.

[18] Robert F. Bales, *Interaction Process Analysis*, Cambridge, Mass.: Addison-Wesley, 1950. This technique is perhaps too complex to be suitable for the analysis of many of the types of interaction occurring within formal organizations; however, the work of some investigators suggests that, with appropriate modifications, it can be adapted for organizational studies.

[19] Eliot D. Chapple, "Measuring Human Relations," *Genetic Psychology Monographs*, 22 (1940), pp. 3–147.

various self-recording devices which may be characterized as indirect observation. Here participants are persuaded to record for the investigator their experiences or their reactions to experiences over a period of time. This operation, too, may be relatively unstructured, as when organization members are asked to keep diaries of their daily activities; or the investigator may enlist the aid of respondents in the collection of more systematic data, for example, a two-week record of all lunch partners. The study of formal organizations affords many opportunities for using direct and indirect observational techniques, which is an important advantage of this field over some others.

Like techniques of observation, interviewing methods may be direct or indirect, systematic or unstructured. The direct unstructured interview, exemplified by asking the manager of an enterprise a wide range of questions about the plant and its operations, is most useful during the early stages of the research, when information is desired on the range and variety of problems confronted by the organization and its members.[20] Once the specific questions and hypotheses that are to guide the research have been formulated, the investigator may choose to utilize the direct interview in a more systematic fashion, asking a list of identical or similar questions of a representative sample of organization members. A good example of this systematic use of the interview is a study in which a sample of 2,451 social scientists from 165 colleges and universities were questioned concerning their views on academic freedom.[21] Finally, the interviewing of organization members may be carried out indirectly by the use of self-administered questionnaires.[22] The advantages of this method are that it is cheap and saves time in comparison with direct interviewing, wherefore a much larger number of persons can be questioned at a given cost. The disadvantages include the relative rigidity of this instrument (all questions will not be understood by, nor be appropriate for, all respondents) and the bias in results introduced by selective returns (disproportionate numbers of uneducated persons, for example, fail to return questionnaires).

The analysis of the organization's documents may be unstructured, as when the analyst surveys the various manuals of operation, charts, rules, and other documents in order to understand its nature and func-

20 For a study based almost entirely on such unstructured interviews with various representatives of management and the union, rank and file workers, and supervisors, see James C. Abegglen, *The Japanese Factory,* Glencoe, Ill.: Free Press, 1958.

21 Paul F. Lazarsfeld and Wagner Thielens, Jr., *The Academic Mind,* Glencoe, Ill.: Free Press, 1958.

22 See, for example, the study of the characteristics and attitudes of a sample of higher civil servants in Egypt conducted by Morroe Berger, *Bureaucracy and Society in Modern Egypt,* Princeton: Princeton University Press, 1957.

tion.[23] Some investigators have attempted a more systematic analysis of organizational documents by using methods of content analysis to establish the frequency with which given themes or topics occur.[24] Performance records, usually plentiful in formal organizations, are another kind of document which may be employed in social research. In the study previously cited, Katz and his associates were able to use company records as the basis for distinguishing between work groups with high and low productivity—a variable fundamental to the purpose of the research.[25] Performance records may also be used to derive indices not contemplated by those who collected them. For example, Blau used organization records to construct an index of competition for each official in two sections of an employment agency.[26] A particular advantage enjoyed by the student of formal organizations is precisely the existence of such documents and performance records; to overlook their significance as an inexpensive, valuable source of data on the policies of the organization and the conduct of its members is to fail to exploit the "natural resources" in this research field.

Types of Research Design. Empirical studies may also be classified on the basis of the research design employed. Three basic designs will be considered here: the sample survey, the controlled experiment, and the field study.[27] As has been pointed out above, sampling surveys are not well suited for studies of organizations; the selection of independent cases is achieved at the cost of tearing individuals from the social matrix in which they are embedded. But this defect is serious only in studies of organizational structure. The survey approach is eminently suited for the study of various other problems of organizational life. To take but a single example, it is appropriate and has been utilized for the study of the careers and attitudes of civil servants, business leaders, and other organization men.[28] In brief, the sample survey is an excellent research design for investigations concerned with the characteristics of the members of organizations but not for studies interested in the characteristics of organizational structure.

[23] Selznick has been particularly imaginative in the unstructured use of documents to gain insights into a new organization and the changes that occurred in it during the first years. Philip Selznick, *TVA and the Grass Roots,* Berkeley: University of California Press, 1949.

[24] For one such approach to organizational documents, see Roy G. Francis and Robert C. Stone, *Service and Procedure in a Bureaucracy,* Minneapolis: University of Minnesota Press, 1956, pp. 51–61.

[25] Katz *et al., op. cit.*

[26] Peter M. Blau, *The Dynamics of Bureaucracy,* Chicago: University of Chicago Press, 1955, pp. 50–55.

[27] For a similar classification of research methods and a full discussion of the several types, see Leon Festinger and Daniel Katz (eds.), *Research Methods in the Behavioral Sciences,* New York: Dryden, 1953, pp. 13–172.

[28] For one such study see W. Lloyd Warner and James C. Abegglen, *Big Business Leaders in America,* New York: Harper, 1955.

At the present stage of our knowledge, complex organizations can neither be placed nor created in the experimental laboratory.[29] Nevertheless, the controlled experiment has a vital contribution to make to the study of organizations, indirectly as well as directly. It provides the ideal model of a scientific research design, which both informs and reveals the limitations of other designs that are less rigorous. The essence of the experimental approach is the comparison of two situations that are identical in all respects except that some factor "A" is introduced in one situation but not in the other. Any differences that consequently develop between the two situations may be attributed to the presence (or absence) of factor "A" since it is the only condition that has been allowed to vary. When experimentation is not possible, in consequence of inability directly to control all factors in the situation, the investigator seeks to approximate the prototype of the controlled experiment by the use of indirect controls through statistical methods or comparative analysis. In the former case, controls are effected through statistical manipulation; in the latter, control is introduced by comparing cases having strategic similarities and differences. Such alternatives, however, are not fully effective since they cannot control all possible sources of variation.

In addition to providing a general model for all scientific research, the experimental method is also of more direct use in the study of social organization. Perhaps its greatest contribution is that it enables one to subject specific hypotheses derived from organizational research to rigorous experimental tests. For example, Blau's tentative conclusion that the prevalence of competition in a group lowers the quality of performance of its members, based on a case study of two sections in a formal organization, receives considerable support from Deutsch's study of the impact of cooperation and competition on task performance in experimental groups.[30] But utilization of the experimental approach by students of formal organization is not limited to such laboratory experiments. In the form of the "field experiment" and the

[29] Interesting attempts are being made, however, to simulate in the laboratory some of the properties of formal organizations, such as structural complexity, formalization, differentiation of units and statuses, integration through hierarchical authority arrangements, and the assignment of tasks where the achievement of one goal interferes with the achievement of others. See Anatol Rapoport, "A Logical Task as a Research Tool in Organization Theory," Mason Haire (ed.), *Modern Organization Theory,* New York: Wiley, 1959, pp. 91–114. The merits of such experimental approaches to the study of formal organizations are discussed in Morris Zelditch, Jr. and Terrence K. Hopkins, "Laboratory Experiments with Organizations," Amitai W. Etzioni (ed.), *Complex Organizations,* New York: Holt, Rinehart and Winston, 1961, pp. 464–478.

[30] Blau, *op cit.*, pp. 49–67; and Morton Deutsch, "An Experimental Study of the Effects of Cooperation and Competition upon Group Process," *Human Relations,* 2 (1949), pp. 199–231.

"natural experiment," this approach has been employed to study organizational patterns in their setting. In the field experiment the investigator actually manipulates some conditions in an existing organization for the purpose of determining causal relations.[31] For example, by actually controlling the amount of participation of various employee groups, Coch and French were able to test experimentally the hypothesis that the greater the participation on the part of these groups in discussing how to cope with changes in working conditions, the less would be their resistance to these changes and the smaller would be the loss in production.[32] In the natural experiment, the investigator does not himself manipulate certain factors within the research setting, but he "opportunistically capitalizes upon some on-going changes and studies their effects in an experimental design."[33] Thus, management may decide to introduce automation in some sections of its plant but not in others. If these sections are similar in other respects, an alert social researcher might grasp the opportunity to observe the impact of technological change on the social relations within the plant by contrasting developments in the automated and the nonautomated sections. In both of these types of situational experiments, unusual caution must be exercised in interpreting findings, since there is always the possibility that the observed changes may be the result of factors that remain undetected by the investigator.

The field study is the typical research design employed in the study of formal organizations. This approach is well adapted for providing an over-all picture of the organization and information about the interdependence of its constituent parts. In the early stages of its development this method was often employed in an impressionistic fashion without the use of rigorous research methods or the presentation of systematic evidence in support of the conclusions. In recent years, however, an increasing number of field studies have utilized carefully designed, systematic research procedures. The field study is particularly hospitable to the combined use of a variety of data-gathering methods, including direct observation, interviewing, and the analysis of documents and records. This advantage is crucial, for it means that the investigator can select from his research repertoire those methods that are the most appropriate for the study of a given problem. Should the field researcher desire data on patterns of social interaction, he can directly observe them; should he require information about the distribution of sentiments, he can inquire about them. A variety of approaches allows him to examine subtle differences which otherwise would escape atten-

[31] See John R. P. French, Jr., "Experiments in Field Settings," Festinger and Katz, *op. cit.*, pp. 98–101.

[32] Lester Coch and John R. P. French, Jr., "Overcoming Resistance to Change," *Human Relations*, 1 (1948), pp. 512–532.

[33] French, *op. cit.*, p. 99.

tion, like that between private feelings and public behavior. Interlocking methods also provide a check on bias by bringing into juxtaposition two or more sets of data on the same problem—say, the observation record of the members at each meeting, and the reports of these members of their attendance at each meeting. Besides, the reliability and validity of various techniques of data collection can be ascertained by comparison. Thus, it can be determined whether the inexpensive reports about meetings furnish data that correlate highly with the more accurate but more expensive data obtained through direct observation.

Although the field study does not approach the controlled experiment in rigor of design and hence in validity of conclusions, the use it permits of a combination of systematic research procedures concentrated on a single object of study can yield data of considerable scientific value. Moreover, its focus on social relations among individuals and groups in natural settings provides data of great importance for the study of organizations—data of a type not obtained by any other design.

Some Practical Problems in the Study of Organizations. We have emphasized that in the field approach to the study of formal organizations, a variety of research techniques may be utilized. We have briefly presented the basic techniques of data collection, and have cited sources where these methods are discussed more fully. We now turn to a general practical problem of field research in an organization, namely, that of the disturbance created by the observer. How can the investigator observe and inquire about social conduct without in the process completely altering that which he wishes to study? This problem is generic to the field study approach, but it is particularly acute in the study of hierarchical organizations.

A procedure frequently employed by the observer to allay suspicion and to minimize the disturbances his presence inevitably produces is to explain at the outset to the staff of the organization who he is and, in a general way, the purpose and methods of his research. This introduction may give him legitimacy in the eyes of the staff and enough prestige so that some doors that might have remained closed to him will be opened. The observer's explanation, however, will probably seem unsatisfactory to many organization members, because he must make his discussion sufficiently vague and general that his major hypotheses will not be revealed, inasmuch as their mere statement could bias the very behavior to be utilized as evidence for their confirmation or rejection. No matter how successful the general explanation and the further explications made to smaller groups and individuals, a certain amount of staff anxiety will probably always accompany the entrance of an observer into the organization. The sources of this anxiety will vary from situation to situation, but their general nature may be outlined. To

begin with, the observer is an outsider and a stranger. He is a figure of some power—frequently of an unknown amount—and will typically have access to powerful people within the organization. Perhaps most importantly, the observer, if he so chooses, can be virtually omnipresent. Having no conflicting duties (as does a supervisor, for example), he can constantly observe the conduct of the group members under study. Merton has pointed out that few if any groups readily accept full and unrestricted observation of their behavior.[34] Finally, the fact that staff members are not certain as to just what the observer's purposes and interests are increases their discomfort. How is one to behave "properly" under observation when the observer's standards for proper behavior are unknown?

It is also likely that misconceptions of the observer's role develop among organization members despite the best efforts of the investigator. The observer may be identified by some as an efficiency engineer or a representative from headquarters sent down to check on the performance of the organization members. Often respondents look on the observer, quite wrongly, as an expert in their field who can help them in the solution of problems. Such misconceptions, whether flattering or disparaging, must be counteracted by the observer's statements and by his behavior; each encounter with organization personnel should be viewed as an opportunity to clarify his own role vis-à-vis the respondent. In short, he must be skilled in the art of "impression-management,"[35] for the observer is also the observed and his behavior is under constant scrutiny from all sides and in all situations.

There are a number of role attributes through which the observer can lessen the disturbance created by his presence and generally increase the reliability of his records, but while each of these solves some problem it tends to raise others. One of these attributes is impartiality: all members of the organization should have approximately equal access to the observer. The avoidance of bias in the selection of cases is a problem in all situations where data are collected; it is a particular problem in a field situation where the "cases" exercise considerable discretion over deciding whether and to what extent they will make themselves available to the observer. Impartiality of access is especially important when two or more different and possibly conflicting groups are being studied, such as various hierarchical levels in an organization.

[34] "Some measure of leeway in conforming to role-expectations is presupposed in all groups. To have to meet the strict requirements of a role at all times, without some degree of deviation, is to experience insufficient allowance for individual differences in capacity and training and for situational exigencies which make strict conformity extremely difficult," Robert K. Merton, *Social Theory and Social Structure* (2d ed.), Glencoe, Ill.: Free Press, 1957, p. 343.

[35] See Erving Goffman, *The Presentation of Self in Everyday Life*, Garden City, N.Y.: Doubleday, 1959, pp. 1–16.

To achieve impartial access the observer can utilize such tactics as often changing his position in the area under study and seeking out the more passive members of the staff. But impartiality will remain only an ideal: some respondents exercise more initiative in contacting the observer than others and are more eager to share with him information otherwise kept confidential, and he cannot afford not to avail himself of these opportunities. Moreover, contacts with certain respondents will prove to be more rewarding—socially as well as in terms of the information received—than contacts with others, and the human observer will inadvertently pay more attention to the former. It should be recognized that during the first few weeks of the study, and to a lesser degree even later, the observer will be an isolate and thus subject to feelings of anxiety and loneliness.[36] Outgoing respondents who are friendly and accepting in these circumstances quite naturally will receive more than their share of his attention.

A second and related attribute of the observer's role is neutrality: his expressions and reactions to all members of the organization's staff should remain neutral and nonjudgmental. Neutrality is essential if an observer is to avoid involvement in the organization with a consequent loss of objectivity and if he is not to inhibit the responses of his informants. But to maintain such neutrality is difficult. Respondents typically attempt to force the observer to express his opinion on various subjects, particularly as he always asks theirs, because his opinion helps them to identify him and to adapt their responses to the kind of relationship they want to maintain with him. In everyday life, such modification of responses by the interpersonal situation occurs continually; but it is precisely what the observer wants to minimize. Yet there is also danger that an impartial neutrality may make the observer seem so dull and colorless that respondents lose all interest in maintaining contact with him. The passively neutral observer may give the impression that he is really not interested in what the respondents have to say, and this impression may inhibit them in his presence. To use an exaggerated example, just think of the effect on the joking banter around a water cooler that an observer would have who never cracked a smile for fear of showing partiality by revealing whose jokes he found funny.

A further attribute of the observer's role is that he is usually not a technical expert in the subject-matter with which respondents deal. Their discovery of his ignorance serves to prove that he is a genuine outside observer and not a "spy" of management, thereby lessening the threat his presence poses. But his technical ignorance may also lower

36 Not only is the observer an isolate but he is constantly confronted by an on-going network of interpersonal relations among the workers observed, which serves to emphasize, by contrast, his own relative isolation.

his prestige in their eyes and thus make some respondents less interested in devoting time to answering his questions.

Another problem confronting the observer in the field is that of maintaining a favorable balance of obligations in interaction with respondents. The observer often finds himself in a position of forever asking and accepting favors from the members of the organization—to let him observe them, to give him information, to explain some document—and his consequent feeling of obligation may impair his effectiveness. But it should be remembered that he has no reason for feeling uneasy about accepting favors from respondents. He offers the staff favors in return: opportunities for demonstrating superior skills before an appreciative audience, for explaining their work and expressing their opinions to an attentive listener, and for making contributions to a scientific study. These are experiences most people enjoy. In addition to these favors inherent in his role, the observer can, and should, take advantage of any occasion to do favors for respondents that do not jeopardize his research role. In our own studies, for example, a number of respondents inquired about the university with which we were affiliated, its course programs, and its degree requirements, and we made it a point to obtain such information for them. Thus the observer can produce a favorable balance of obligations which motivates the members of the organization to cooperate with the research. But it should be mentioned that there is the opposite danger of having respondents too eager to cooperate, since this situation may also bias their verbal statements as well as their overt behavior.

Several situational factors, fortunately, lessen the disturbance created by the presence of the observer. The main one is the network of relations among the members of the organization. Once respondents have become accustomed to the investigator and, particularly, have accepted him as a genuine outside observer, his presence changes little in their situation. Each respondent has his familiar duties and is surrounded by his colleagues and his superiors, all of whom have developed distinct expectations of him and specific relations with him. The constraints exerted upon the individual by the expectations of other members of the organization and his relations with them appear generally to be too great to let him permit the presence of the outsider to alter his conduct very much.[37] The influence of interpersonal relations is most pronounced in the very areas where the observer's presence would otherwise distort conduct most. Members of an organization typically seek to conceal conflicts from outsiders, but if the observer happens to be present when a conflict arises, the emotional involve-

[37] Note that the same safeguards do not operate in the interview situation. Here the person is temporarily separated from his social context and may distort his comments in any way he sees fit without fear of retaliation from his peer group.

ment of participants in the argument makes it difficult for them to dissimulate because he is there. Another aid to the observer in collecting reliable data is what may be termed the bias-correcting action of respondents. Most organization members will be anxious to make sure that the observer obtains what they consider to be a "true" picture of their activities and opinions. Thus, if one member in the presence of his fellows makes a statement with which they do not agree, they will often challenge it. Since the observer cannot easily challenge the statements of his respondents, these workers perform a valuable service for him by uncovering misleading statements or areas of disagreement between organization members.

In short, the observer would do well to keep constantly in mind that he is playing a social role in a social situation and to adapt his role to his research objectives. This recommendation does not imply, however, that he should be manipulative, but rather the contrary; attempts to simulate and dissimulate are too easily detected in a group situation to be useful strategies for the observer.

The Comparative Approach

The objective of this book is theoretical: to suggest tentative generalizations that help to explain the structure and dynamics of organizations. The emphasis in this book, however, is on the analysis of research findings. Since valid generalizations must be both derived from and tested in empirical research, we shall be largely concerned with empirical studies rather than with general discussions of administrative principles or criticisms of the bureaucratic tendencies in American life. The major empirical sources to be used are: (1) reports of research on various organizations; (2) experiments and other studies from the literature having bearing upon some organizational problem; and (3) our own investigations, especially our research on two welfare agencies.

The approach of this book, finally, is comparative. By "comparative" we mean that the importance of comparing different organizations for generalizing about them informs our thinking throughout the analysis. We cannot claim more than that, inasmuch as most research on this subject that has been reported in the literature consists of case studies. Indeed, in view of this situation, our attempt to be comparative may seem hazardous if not foolhardy. Nevertheless, we believe that the fruits to be attained by even partial success justify the risks involved. How do we propose to proceed with this comparative study, and from where shall the data come? First, it will be possible to attempt to make comparisons between several of the case studies of formal organizations in the literature. Second, we shall draw on those studies that make

internal comparisons, for example, of one department with another. Third, we can make use of the few studies that have attempted to make systematic comparisons of several organizations, such as Udy's analysis of the organization of work in primitive societies.[38] Finally, we shall draw extensively on our own separate studies of two welfare agencies, which are described in the appendix. Since we shall have many occasions to refer to these two studies in the following chapters, it should be noted that the fact that these organizations are public welfare departments is incidental to our purpose here; we are interested not in public assistance but in formal organization.[39] The intent of our extensive use of these two studies is to provide some continuity in the illustrative materials throughout the book.

38 Stanley H. Udy, *The Organization of Work*, New Haven: Human Relations Area Files, 1959.

39 For a general characterization of social welfare agencies in this country—both with regard to program content and structural characteristics—see Harold L. Wilensky and Charles N. Lebeaux, *Industrial Society and Social Welfare*, New York: Russell Sage Foundation, 1958.

CHAPTER TWO

The Nature and Types of Formal Organizations

To clarify the nature and characteristics of formal organizations, we shall first deal with the major conceptions advanced by organization theorists and then discuss the various forms that such organizations may assume. We begin by presenting Weber's classical theories of authority and of bureaucratic organization. Following a critical analysis of Weber's conception, we shall briefly deal with two other general theoretical treatments of formal organizations. With regard to types of organizations, our main purpose is to suggest some of the dimensions along which classification may occur and to illustrate the utility of one approach for the development of a classificatory scheme.

Theoretical Concepts

The Concept of Authority. Max Weber's perceptive and incisive theoretical analysis of the principles of bureaucracy is undoubtedly the most important general statement on formal organization. Since its publication in *Wirtschaft und Gesellschaft* about 40 years ago, it has had a profound influence on almost all subsequent thinking and research in the field. Weber analyzes formal organizations as part of his theory of authority structures, or systems of legitimate social control; we shall follow his procedure and begin with a discussion of the concept of authority.

Authority must be distinguished from other forms of social influence—from power, on the one hand, and from persuasion and other kinds of personal influence, on the other. Weber defines power as "the probability that one actor within a social relationship will be in a position to carry out his own will despite resistance."[1] Power, as used by Weber, is a very general and comprehensive concept. It includes control through the use or the threat of physical coercion (as exercised by the stick-up man or by the rapist) and it includes the control of one who can manipulate conditions in such a way that others are forced to

1 Max Weber, *The Theory of Social and Economic Organization*, A. M. Henderson and Talcott Parsons (trans.) and Talcott Parsons (ed.), Glencoe, Ill.: Free Press and Falcon's Wing Press, 1947, p. 152.

act in his interests rather than their own (as exemplified by a monopoly or by the one-company town).

The stick-up man and the monopolist exercise power, but they do not exercise authority, since the essence of the latter is that directives issued by the one in control are *voluntarily* obeyed. Weber defines authority as "the probability that certain specific commands (or all commands) from a given source will be obeyed by a given group of persons."[2] The group willingly obeys because its members consider it legitimate for this source to control them. The source of authority may be a person or it may be an impersonal institution, such as a system of laws. The fact that one individual voluntarily permits another to influence his behavior, however, is not necessarily indicative of an authority relation. There are also other types of personal influence, such as persuasion. In persuasion, one person lets the arguments of another person influence his decisions or actions. Persuasion is distinguished from authority inasmuch as the latter involves an a priori suspension of the first person's judgment, obviating the need for persuasion. In an authority relation the subordinate "holds in abeyance his own critical faculties for choosing between alternatives and uses the formal criterion of the receipt of a command or signal as his basis for choice."[3]

Two criteria of authority, then, are voluntary compliance with legitimate commands and suspension of judgment in advance of command. In concrete situations of social control, however, it is often difficult to determine whether these two criteria are actually met. Take the extreme case of the slave driver whose coercive power rested on his whip. He did not use his whip continually, since the very knowledge of the possibility of its use sufficed to make the slaves obey his orders. While few would classify this case as one of voluntary compliance, it differs only in degree from the case of the employer whose economic dominance over his employees motivates them to carry out official directives; yet, the latter relationship is usually considered to be one of authority. Again, when a person has often persuaded another in the past, his opinion may well carry so much weight that the second person lends a willing ear to his arguments and requires hardly any persuasion to be influenced. Does the first person have authority over the other, or not?

A major reason why there are so many borderline cases is that other forms of social control often develop into authority. For this change to occur, however, another social condition must be met, and this condition provides a final and basic distinguishing criterion for authority. The condition is that a value orientation must arise that

[2] *Ibid.*, p. 324.

[3] Herbert A. Simon, *Administrative Behavior* (2d ed.), New York: Macmillan, 1957, pp. 126–127.

defines the exercise of social control as legitimate, and this orientation can arise only in a group context. A single individual forced to comply with the commands of another may seek to adapt to this situation by rationalizing that he really wants to be guided by the other's directives. That this attitude is a mere rationalization would be indicated by the fact that once the coercive power were withdrawn he would not continue to comply. If, however, an entire group finds itself in the same situation, and if its members share the beliefs that it is good and right and, indeed, in their best interest to obey, the rationalizations of individuals become transformed into a common value orientation. For group agreement and approval of what is right constitute a social value that validates the beliefs of individuals and thereby makes them "really" right. Given the development of social norms that certain orders of superiors ought to be obeyed, the members of the group will enforce compliance with these orders as part of their enforcement of conformity to group norms. The group's demand that orders of the superior be obeyed makes such obedience partly independent of his coercive power or persuasive influence over individual subordinates and thus transforms these other kinds of social control into authority.

A fundamental characteristic of authority, therefore, is that the willingness of subordinates to suspend their own judgment in advance and follow the directives of the superior results largely from social constraints exerted by the collectivity of subordinates and not primarily from the influences the superior himself can bring to bear upon them.[4] Such social constraints are not characteristic of coercive powers, persuasion, or other types of personal influence. If a boy is in love with a girl, for example, he will be eager to do what she wants in order to make his company more attractive to her. Similarly, if a person depends on a colleague for advice, he will feel obligated to accede to this colleague's wishes. In both of these cases, one person's dependence on another for certain rewards motivates him to comply with the other's requests in exchange for obtaining these rewards. But it would not be correct to classify such patterns of personal influence, which are rooted in exchange processes, as authority, even though compliance is voluntary, because there is no social collectivity whose norms require compliance with the directives of the superordinate person in the relation.

Authority relations can develop only in a group or larger collectivity, and not in isolated pairs, because only group values can legitimate the exercise of social control and only group norms can serve as an independent basis for enforcing the pattern of compliance. Once an authority structure has become institutionalized, however, it can find

[4] The compliance of subordinates in authority relations is voluntary but not independent of social constraints. It is as voluntary as is our custom of wearing shoes on the street.

expression in apparently isolated pair relationships. Thus, a father exercises authority over his son, even though there is only one child in the family. The father's influence does not entirely depend on his superior power, or his success in persuasion, or his willingness to exchange favors for obedience, because culturally defined role expectations obligate the son to obey his father; and members of the community, such as teachers and neighbors, help to enforce these obligations. To be sure, other forms of influence often exist side by side with authority and extend the influence of superiors. Pure authority relationships are analytical abstractions that are found rarely, if at all, in concrete situations, but this rarity makes the analytical distinction between them and other forms of influence no less important.

Weber's Types of Authority. We have seen that authority exists only when a common value orientation in a collectivity legitimates the exercise of control as appropriate and proper. The types of legitimating beliefs that support the exercise of control vary, and Weber uses these differences as the basis for distinguishing three types of authority.[5]

The first type is authority legitimated by the sanctity of tradition. In "traditional" authority, the present social order is viewed as sacred, eternal, and inviolable. The ruling person or group, usually defined by heredity, is thought to have been ordained by supernatural powers to rule over the rest. The subjects are bound to their ruler by traditional feelings of personal loyalty and other cultural beliefs about the social order that reinforce his position, such as the concept of the divine right of kings. Absolute monarchies exemplify traditional authority, as does the patriarchal family or a shop under a paternalistic boss. The ruler's power is great under this system. Although it is limited by the traditions that legitimate it, this restriction is not severe, since a certain amount of arbitrariness on the part of the ruler is often part of the tradition and since a precedent in the past usually can be found to justify his new commands. The need to turn to past traditions for legitimation of present actions, however, sets this type of authority apart from others. In general, traditional authority tends to perpetuate the existing social order and is ill suited for adaptation to social change; indeed, change undermines its very foundation.

The values that legitimate a "charismatic" authority, Weber's second type, define a leader and his mission as being inspired by divine or supernatural powers. The leader, in effect, heads a new social movement, and his followers or disciples are converts to a new cause. There

[5] Weber, *op. cit.*, pp. 324–386. Note the following comment by Bendix: "In Weber's view beliefs in the legitimacy of a system of domination are not merely philosophical matters. They can contribute to the stability of an authority relationship, and they indicate very real differences between systems of domination." Reinhard Bendix, *Max Weber,* Garden City, N.Y.: Doubleday, 1960, p. 297.

is a sense of being "called" to spread the new gospel, a sense of rejecting the past and heralding the future as symbolized in Christ's words, "It is written . . . , but I say unto you . . ." Devotion to the leader and a conviction that his actions embody the newly adopted ideals are the source of the group's willing obedience to his commands. Charismatic leaders may appear in almost any arena of social life, as prophets in religion, demagogues in politics, and heroes in battle. Charismatic authority generally functions as a revolutionary force, rejecting the traditional values and rebelling against the established order.

Initially, charismatic movements tend to be anarchistic and eschew even internal organization. Revolutionary ideals and the urgency of the mission make the charismatic leader and his followers contemptuous of anything associated with routines or "business as usual." No confining organization or rigid rules must be allowed to fetter the leader's inspiration, and the devotion to the sacred mission must not be profaned by mundane economic considerations. But it seems axiomatic that the demands of reality—for some routine, some organization, some stable means of economic support—cannot be ignored indefinitely. If it is to persist, the movement must become organized. Accordingly, the members' interest in perpetuating the movement constrains them to overcome their distaste for formal routines and to develop a formal organization. The necessity for organization becomes particularly apparent with the impending death of the charismatic leader. This event confronts the movement with the dangers of collapsing due to lack of inspired guidance or of being torn asunder by struggles over succession unless a regular procedure has been developed for transferring the leader's mantle to a legitimate heir. Charismatic movements are inherently unstable, since they are linked to the life of one man; to endure, they must take on some of the characteristics of that mundane world which initially they arose to combat and change. This process of "the routinization of charisma," as Weber called it, transforms charismatic authority into a different type; it may crystallize into a traditional system or, more often, become bureaucratized into the legal authority of a formal organization.[6]

Weber's third type, "legal" authority, is legitimated by a belief in the supremacy of the law. This type assumes the existence of a formally established body of social norms designed to organize conduct for the rational pursuit of specified goals. In such a system obedience is owed not to a person—whether a traditional chief or a charismatic leader—but to a set of impersonal principles. These principles include the re-

[6] Dostoyevsky's "Grand Inquisitor" graphically deals with the conflict between charisma and organization in the form of a fictional discussion between Christ and a Cardinal of the Church. Fyodor Dostoyevsky, *The Brothers Karamazov,* New York: Random House, 1937, pp. 292–314.

quirement to follow directives originating from an office superior to one's own, regardless of who occupies the higher office. All formal organizations—the government, a factory, the army, a welfare agency—are examples of legal authority structures. The epigram for this type might well be: "a government of laws, not of men." Although superior officials command the obedience of their subordinates, they in turn are subject to the authority of the same body of impersonal regulations, and their authority is accordingly limited.

Each of the three types of authority creates an appropriate staff of assistants. Thus, under the traditional form the feudal lord has his vassals and the king his court and retainers; the charismatic prophet has his disciples; and under the legal form, a staff of administrative officials becomes established and rationally organized. The distinctive characteristic of the latter type is that this bureaucratically organized administrative staff is capable of tremendous expansion and often becomes the dominant feature of legal authority systems.

Weber's Theory of Bureaucracy. Almost all modern administrative organizations (as well as some ancient ones) are bureaucratically organized. Weber enumerates the distinctive characteristics of this type of organization in the following way:[7]

(1) Organization tasks are distributed among the various positions as official duties. Implied is a clear-cut division of labor among positions which makes possible a high degree of specialization. Specialization, in turn, promotes expertness among the staff, both directly and by enabling the organization to hire employees on the basis of their technical qualifications.

(2) The positions or offices are organized into a hierarchical authority structure. In the usual case this hierarchy takes on the shape of a pyramid wherein each official is responsible for his subordinates' decisions and actions as well as his own to the superior above him in the pyramid and wherein each official has authority over the officials under him. The scope of authority of superiors over subordinates is clearly circumscribed.

(3) A formally established system of rules and regulations governs official decisions and actions. In principle, the operations in such administrative organizations involve the application of these general regulations to particular cases. The regulations insure the uniformity of operations and, together with the authority structure, make possible the coordination of the various activities. They also provide for continuity in operations regardless of changes of personnel, thus promoting a stability lacking, as we have seen, in charismatic movements.

[7] Weber's discussion of these characteristics may be found in H. H. Gerth and C. Wright Mills (trans. and eds.), *From Max Weber: Essays in Sociology*, New York: Oxford University Press, 1946, pp. 196–204; and in Weber, *op. cit.*, pp. 329–336.

(4) Officials are expected to assume an impersonal orientation in their contacts with clients and with other officials. Clients are to be treated as cases, the officials being expected to disregard all personal considerations and to maintain complete emotional detachment, and subordinates are to be treated in a similar impersonal fashion. The social distance between hierarchical levels and that between officials and their clients is intended to foster such formality. Impersonal detachment is designed to prevent the personal feelings of officials from distorting their rational judgment in carrying out their duties.

(5) Employment by the organization constitutes a career for officials. Typically an official is a full-time employee and looks forward to a lifelong career in the agency. Employment is based on the technical qualifications of the candidate rather than on political, family, or other connections. Usually such qualifications are tested by examination or by certificates that demonstrate the candidate's educational attainment —college degrees for example. Such educational qualifications create a certain amount of class homogeneity among officials, since relatively few persons of working-class origins have college degrees, although their number is increasing. Officials are appointed to positions, not elected, and thus are dependent on superiors in the organization rather than on a body of constituents. After a trial period officials gain tenure of position and are protected against arbitrary dismissal. Remuneration is in the form of a salary, and pensions are provided after retirement. Career advancements are "according to seniority or to achievement, or both."[8]

In Weber's view, these organizing principles maximize rational decision-making and administrative efficiency. Bureaucracy, according to him, is the most efficient form of administrative organization, because experts with much experience are best qualified to make technically correct decisions, and because disciplined performance governed by abstract rules and coordinated by the authority hierarchy fosters a rational and consistent pursuit of organizational objectives.

Refinements of Weber's Conceptions. Weber analyzes bureaucratic organizations not empirically but as an ideal type. He does not characterize the "average" administrative organization; rather, he seeks to bring together those characteristics that are distinctive of this type. Just as we can imagine physicians constructing a model of the perfectly healthy man, so Weber attempts to characterize a perfectly bureaucratized organization. A question which remains to be answered is, what is the criterion of perfection employed by Weber?

Weber's construct of ideal type is an admixture of a conceptual scheme and a set of hypotheses. As a conceptual scheme it defines the

[8] *Ibid.*, p. 334.

phenomenon to be studied. His conception calls attention to what he considers the key elements of bureaucratic organization—those essential for understanding such organizations. Weber provides a definition of the concept of bureaucracy by pointing to these characteristics. He says in effect: bureaucracies are those organizations that exhibit the following combination of characteristics. Such conceptual schemes provide important frameworks for analysis and research, although they themselves are not subject to empirical testing. They are neither correct nor incorrect, only more or less useful in guiding scientific investigations.

But in addition to its conceptual elements, the ideal type contains a series of hypotheses. Weber suggests that many of the characteristics attributed to bureaucracies are interrelated in particular ways; for example, specialization is said to promote expertness, the authority structure and the existence of formal rules are assumed to make vital contributions to the coordination of activities, and detachment is held to increase rationality. Further, Weber states that these characteristics, and, specifically, their combination function to maximize administrative efficiency. A careful reading of Weber indicates that he tends to view elements as "bureaucratic" to the extent that they contribute to administrative efficiency. This contribution to efficiency appears to be the criterion of "perfect" embodied in his ideal type. However, whether or not each of these elements, or their combination, enhances administrative efficiency is not a matter for definition; these are questions of fact—hypotheses subject to empirical testing.

To exploit Weber's insightful theoretical analysis, it is necessary, in our opinion, to discard his misleading concept of the ideal type and to distinguish explicitly between the conceptual scheme and the hypotheses. The latter can then be tested and refined rather than left as mere impressionistic assertions. We can ask, for example: does tenure promote efficiency? under what conditions does it have this effect, and under what conditions does it not? Only in this way can we hope to progress beyond Weber's insights to the building of systematic theory.

Weber has often been criticized for presenting an idealized conception of bureaucracy. His statements appear to entail an implicit functional scheme; again and again he addresses himself to the problem of how a given element contributes to the strength and effective functioning of the organization. What is missing is a similar systematic attempt to isolate the *dysfunctions* of the various elements discussed[9] and to examine the conflicts that arise between the elements comprising the system. Thus, even if it is true that the hierarchy of authority promotes discipline and makes possible the coordination of activities, does it not

[9] Merton has called for this balanced approach in his paradigm for functional analysis. Robert K. Merton, *Social Theory and Social Structure* (2d ed.), Glencoe, Ill.: Free Press, pp. 50–54.

also discourage subordinates from accepting responsibility? Or, granted that promotion should be based on objective criteria rather than on personal considerations or family connections, which of the two major criteria is to be selected—seniority or merit? When questions such as these are raised, it is seen that Weber's one-sided concern with the functions of bureaucratic institutions blinds him to some of the most fundamental problems in the study of formal organizations.[10]

Another criticism that has been advanced against Weber's analysis is that he is preoccupied with the formally instituted aspects of bureaucracies and ignores the informal relations and unofficial patterns which develop in formal organizations. Selznick has emphasized that the formal structure is only one aspect of the actual social structure and that organizational members interact as whole persons and not merely in terms of the formal roles they occupy.[11] Many empirical studies demonstrate that friendship patterns, unofficial exchange systems, and "natural" leaders arise to modify the formal arrangements.[12] Weber's conceptual scheme, by concentrating on the officially instituted aspects of bureaucracies, neglects the ways in which these are modified by informal patterns and thus excludes from analysis the most dynamic aspects of formal organizations.

Finally, Parsons and Gouldner have called attention to an implicit contradiction in Weber's conception of bureaucracy; in Gouldner's words, "On the one side, it was administration based on expertise; while on the other, it was administration based on discipline."[13] By emphasizing both expert judgment resting on technical knowledge and disciplined compliance with directives of superiors as the basis for bureaucratic decisions, Weber implies that there is no conflict between these two principles; that is, he implicitly assumes that in every disagreement between superior and subordinate, the superior's judgment is also the better judgment in terms of technical expertise. This is not a realistic assumption. Executives in complex organizations are not merely occasionally but typically less qualified to make expert technical judgments than their professional subordinates, since they cannot possibly be the

10 Defenders of Weber will argue that he did note certain conflicts or dilemmas inherent in bureaucratic structures. The point here is, however, that he did not subject these dysfunctions to the same kind of systematic analysis that he furnished for the functions of various characteristics for bureaucratic efficiency.

11 Philip Selznick, "Foundations of the Theory of Organization," *American Sociological Review,* 13 (1948), pp. 25–35.

12 See, for example, Charles H. Page, "Bureaucracy's Other Face," *Social Forces,* 25 (1946), pp. 88–94, and Ralph H. Turner, "The Navy Disbursing Officer as a Bureaucrat," *American Sociological Review,* 12 (1947), pp. 342–348.

13 Alvin W. Gouldner, *Patterns of Industrial Bureaucracy,* Glencoe, Ill.: Free Press, 1954, p. 22. See also Talcott Parsons' "Introduction," in Weber, *op. cit.,* pp. 58–60, footnote 4.

leading expert in each of the specialties under their jurisdiction. Often, indeed, they are experts not in any of these specialties but in administration. Administrative considerations, moreover, tend to conflict with technical professional considerations. Hence, the judgment of superiors, who are concerned with administrative problems, will recurrently differ from the judgment of their professional subordinates, who are concerned with technical problems. Generally, one of the central issues in contemporary professional organizations, which will occupy us repeatedly in later chapters, is the conflict between disciplined compliance with administrative procedures and adherence to professional standards in the performance of duties.

Weber's pioneering analysis of bureaucracy has stimulated much further analysis and research on formal organizations, and these studies make it possible critically to review and to refine some of his theoretical concepts. Our criticisms of parts of his discussion on the basis of more recent work in sociology are not intended to disparage the outstanding contribution Weber has made to this field.

Other Conceptions of Formal Organizations. Weber is not the only social scientist who has attempted to develop a theoretical framework for the analysis of formal organizations. The approach of two other students of social organization will be briefly considered here, although our list could easily be extended.

Herbert Simon conceives of administrative organizations primarily as decision-making structures.[14] He has characterized his own focus in the following passage:

> What is a scientifically relevant description of an organization? It is a description that, so far as possible, designates for each person in the organization what decisions that person makes, and the influence to which he is subject in making each of these decisions.[15]

Effective administration, according to Simon, requires rational decision-making; decisions are rational when they select the best alternative for reaching a goal. But administrative decisions are highly complex, and rationality is limited for various reasons: all the consequences that follow from a given course of action cannot be anticipated; the consequences of action lie in the future and thus are difficult to evaluate realistically; and rationality requires a choice among all possible alternatives, but many of these will never even come to mind and so will not be considered. In short, individuals are not capable of making complex decisions rationally. The function of the organization is to limit the scope of the decisions that each member must make; only in

14 See Simon, *op. cit.*, pp. 1–11, 45–78, *et passim*.
15 *Ibid.*, p. 37.

this way can rationality be approached.[16] How does the organization accomplish this feat? First, it defines the responsibilities of each official, thus supplying him with goals to guide his decisions. Second, it sets up the mechanisms, such as formal rules, information channels, and training programs that help to narrow the range of alternatives the official must consider before making his decisions.

This conclusion may be somewhat amplified. Decisions, says Simon, are based on two types of premises. There are factual premises, which are subject to empirical testing in order to ascertain their truth or falsity, and value premises, which are not subject to such tests since they are concerned not with what "is" but with what "ought" to be, what is good or preferable. The former have to do with the choice of means, the latter with the choice of ends. Rational behavior may be viewed as consisting of means-ends chains. Given certain ends, appropriate means are selected for their attainment; but once reached, the ends often become the means for the attainment of further ends, and so forth. For example, a student studies in order to obtain a high mark in a course; he desires a high mark because this will help him attain a high over-all grade average; he desires a high grade average in order to be able to graduate from college; college graduation, in turn, is only a means to the end of getting a desirable job; and so on. The important point about organizational behavior is that the hierarchical structure permits all decisions, except those defining ultimate objectives, to rest on factual rather than on value premises, that is, to be decisions about means rather than ends. Once the objectives of the organization are formally established, the hierarchical organization of responsibilities serves as a framework for means-ends chains—specifying for each official the ends of his tasks and thus confining his duties to the selection of the best means for achieving these ends. To illustrate, the duty of the top manager may be described as finding effective ways for accomplishing the established objectives of the organization; the means he employs for this purpose are work assignments to his department heads. These assignments provide each department with its objectives, which the head seeks to attain by means of assigning responsibilities to the supervisors under him, and so on down the line. The ends of every member of the organization are defined by the directives of his superior, and his responsibility is primarily to decide on the best means for attaining these ends. In other words, each official in the hierarchy

16 Rationality is always only approached, never achieved. The organization member must be willing to forgo a search for the best of all possible alternatives and be content with finding a satisfactory one; in Simon's language, administrators must "satisfice [sic] because they have not the wits to maximize." *Ibid.*, p. xxiv· see also James. G. March and Herbert A. Simon, *Organizations,* New York: Wiley, 1958, pp. 140–141, 169.

has his value premises supplied by his superior; besides, his search for alternative means is narrowed by procedural regulations. The combination of these two limits, according to Simon, permits rational decision-making in an organization.

Simon's suggestive conception of administration as a decision-making structure deals largely with the effects of the formal blueprint on decision-making and does not include a systematic analysis of those interpersonal processes that are not part of the formal structure. Our discussion, in contrast, will be particularly concerned with the significance of these informal interpersonal influences for decision-making. Another limitation of Simon's analysis is that he directs all his efforts to explaining how the various conditions in the organization—the hierarchy, the communication system, training programs—influence rational decision-making and omits consideration of their influence on one another. Such a specific focus on choice behavior as the sole dependent variable makes it impossible systematically to analyze social structure, since it reduces all problems of social structure to sociopsychological problems; that is, all questions of "what produces these characteristics of the organization?" are turned into a concern with "what produces this behavior of the members in the organization?"

Talcott Parsons provides yet another conception of formal organization in the recent application of his general theoretical framework for the study of social systems to such organizations.[17] According to Parsons' schema, all social systems must solve four basic problems: (1) adaptation: the accommodation of the system to the reality demands of the environment coupled with the active transformation of the external situation; (2) goal achievement: the defining of objectives and the mobilization of resources to attain them; (3) integration: establishing and organizing a set of relations among the member units of the system that serve to coordinate and unify them into a single entity; and (4) latency: the maintenance over time of the system's motivational and cultural patterns.[18] This scheme has sufficient generality to be applicable to all social systems. For example, if our focus is the society, then the economy may be said to function as the subsystem meeting the problems of adaptation faced by the society. Formal organizations, although they serve different specific functions, are part of the goal attainment subsystem of the larger society. Parsons thus views formal organizations as a major mechanism in modern societies for mobilizing power in the interest of achieving collective objectives.

But each formal organization may also be viewed as a social system in its own right that must possess its own set of subsystems concerned

[17] Talcott Parsons, *Structure and Process in Modern Societies*, Glencoe, Ill.: Free Press, 1960, pp. 16–96.

[18] See Talcott Parsons *et al.*, *Working Papers in the Theory of Action*, Glencoe, Ill.: Free Press, 1953, pp. 183–186.